The Department of Computer Science is a part of the Faculty of Information & Communication Technology (FICT) and is officially accredited by the National Computing Education Accreditation Council (NCEAC). Founded in 2002 as one of the university's pioneer undergraduate programs, it later expanded to include an MS degree program in 2006 and a PhD program in 2014. The core computational focus of the department is to provide foundational and technical knowledge in areas including computer communication, networking, security, software development, database management systems, data sciences, and core programming. Graduates often transition into professional roles such as software engineers, data scientists, system administrators, and IT managers within industrial and governmental laboratories. The degree is also designed to serve as a multidisciplinary launchpad for careers in fields such as digital medicine, law, education, life sciences, and the humanities. The department's vision is to offer a comprehensive computational education environment that inspires curiosity, creativity, and innovation to enhance society in a positive and meaningful way. Program Educational Objectives state that graduates are expected to responsibly practice systematic, disciplined, and quantifiable approaches to solving complex problems across computing and allied disciplines. Furthermore, graduates should utilize their skills and knowledge to serve both industry and academia to positively impact the socio-economic development of Pakistan and demonstrate sustained learning and adaptability to evolving technological fields through continuous professional development and self-study.

For the Bachelor of Science in Computer Science (BSCS), admission requirements include an F.Sc in Pre-Engineering or ICS with Mathematics and Physics from a recognized board or an equivalent qualification with at least 60% marks, or a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks. Additionally, candidates are required to clear an entrance exam administered by NTS. The degree requirements for the BSCS program consist of a total of 133 to 134 credit hours and 37 to 43 courses, while maintaining a minimum academic standing of a CGPA of 2.0 or higher. In the curriculum plan, the credit hour listings follow a format of theory plus lab. The 1st semester includes Introduction to ICT & Lab (CS-110/CS-110L) with 2+1 credit hours and no prerequisites, Programming Fundamentals & Lab (CS-114/CS-114L) with 3+1 credit hours and no prerequisites, and English Composition & Comprehension (HUM-175) with 3+0 credit hours and no prerequisites. Also in the 1st semester are Calculus & Analytical Geometry (MATHP-105) with 3+0 credit hours and no prerequisites, Applied Physics & Lab (PHY-205 / PHY-205L) with 2+1 credit hours and no prerequisites, and Islamic Studies (HUM-101) with 2+0 credit hours and no prerequisites. The 2nd semester courses are Object Oriented Programming & Lab (CS-212/L) with 3+1 credit hours and Programming Fundamentals as a prerequisite, Communication & Presentation Skills (Hum-268) with 3+0 credit hours and English Composition & Comprehension as a prerequisite, and Multivariable Calculus (CS Supporting I) with 3+0 credit hours and Calculus & Analytical Geometry as a prerequisite. Rounding out the 2nd semester are Digital Logic & Design & Lab (EE-102/L) with 3+1 credit hours and no prerequisites, and Discrete Structures (MATHA-234) with 3+0 credit hours and Applied Physics as a prerequisite.

The 3rd semester of the BSCS program features Data Structure & Algorithms & Lab (CS-214/L) with 3+1 credit hours and Object Oriented Programming as a prerequisite, Software Engineering (CS-334) with 3+0 credit hours and no prerequisites, and Computer Organization & Assembly Language (CE-213/L) with 3+1 credit hours and Digital Logic & Design as a prerequisite. Also included are Introduction to Psychology (University Elective I) with 3+0 credit hours and no prerequisites, and Linear Algebra (MATHP-111) with 3+0 credit hours and no prerequisites. The 4th semester includes Operating Systems & Lab (CS-341/L) with 3+1 credit hours and Data Structure & Algorithms as a prerequisite, Computer Networks & Lab (TE-307/L) with 3+1 credit hours and no prerequisites, and Theory of Automata (CS-321) with 3+0 credit hours and no prerequisites. This semester also contains Numerical Computing (CS Supporting II) with 3+0 credit hours and no prerequisites, and Human Resource Management (University Elective II) with 3+0 credit hours and no prerequisites. The 5th semester starts with Technical & Business Writing (HUM-266) with 3+0 credit hours and no prerequisites, followed by Compiler Construction (CS-421) with 3+0 credit hours and Theory of Automata as a prerequisite. Other 5th semester courses include Artificial Intelligence & Lab (CS-412/L) with 3+1 credit hours and Discrete Structures as a prerequisite, Database Systems & Lab (CS-332/L) with 3+1 credit hours and Data Structure and Algorithm as a prerequisite, and Probability and Statistics (STAT-101) with 3+0 credit hours and no prerequisites.

The 6th semester of the BSCS program consists of Design & Analysis of Algorithm (CS-424) with 3+0 credit hours and Data Structure and Algorithm as a prerequisite, Introduction to Big Data (CS Elective I) with 2+1 credit hours and no prerequisites, and Machine Learning (CS Elective II) with 2+1 credit hours and no prerequisites. Also in the 6th semester are Theory of Programming Languages (CS Supporting III) with 2+1 credit hours and no prerequisites, Marketing (University Elective III) with 3+0 credit hours and no prerequisites, and Social Service (University Elective IV) which is a 0+0 credit hour pass or fail course with no prerequisites. The 7th semester includes Final Year Project-I (TBA) with 0+3 credit hours and Software Engineering and Technical Writing as prerequisites, Professional Practices (HUM-309) with 3+0 credit hours and no prerequisites, and Parallel & Distributed Computing (TBA) with 3+0 credit hours and Operating Systems as a prerequisite. Rounding out the 7th semester are Information Security (TBA) with 3+0 credit hours and no prerequisites, and Cloud Computing (CS Elective III) with 3+0 credit hours and no prerequisites. The 8th semester features Final Year Project-II (TBA) with 0+3 credit hours and Final Year Project-I as a prerequisite, Pakistan Studies (HUM-102) with 2+0 credit hours and no prerequisites, and Advanced DBMS (CS Elective IV) with 3+0 credit hours and no prerequisites. The final courses in the 8th semester are Cryptography (CS Elective V) with 3+0 credit hours and no prerequisites, and Wireless Networks (University Elective V) with 3+0 credit hours and no prerequisites.

For the Master of Science (MS) in Computer Science, admission requires 16 years of education or an equivalent 4-year degree in a relevant discipline from an HEC-recognized university, with at least 60% marks or a minimum CGPA of 2.5 out of 4.0. Candidates must also achieve a cumulative score of at least 50% on an entry test conducted by the university or an authorized testing agency. The MS degree requirements include a total of 30 credit hours distributed across 8 to 10 graduate-level courses, and students must maintain a minimum CGPA of 2.5. Admission to the Doctor of Philosophy (PhD) in Computer Science requires an 18-year master's degree featuring at least 30 credit hours, comprised of 24 credit hours of graduate coursework and 6 credit hours of research thesis, from an HEC-recognized institution in a relevant field. PhD candidates must have at least a 1st Division pass under the annual system or a minimum CGPA of 3.0 out of 4.0. Before admission is confirmed, candidates must pass the standard international GRE Subject Test with a minimum 60th percentile score or an authorized GAT Subject Test with at least 60% marks. The PhD degree requirements involve 18 total credit hours from 6 to 8 advanced doctoral courses and mandatory dissertation work, with a minimum academic standing of a 2.5 CGPA.